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Batch : B3

ASSIGNMENT No: 4

TITLE: Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Theory

Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required.

Code

```
public class Wrapper{

    public static void main(String[] args) {

        // Autoboxing
        int num=509;
        Integer intObj=num;

        double value= 20.67;
        Double doubleObj=value;

        // Unboxing
        int primitiveInt=intObj;
        double primitiveDouble=doubleObj;

        //String to Wrapper
        Integer number= Integer.valueOf("50");
        Double decimal= Double.valueOf("100.38");
```

```

//Wrapper to String
String str1 = intObj.toString();
String str2 = doubleObj.toString();

//Basic Operations

int sum = intObj + number;
double product = doubleObj * decimal;

System.out.println("primitive Int: "+primitiveInt );
System.out.println("primitive Double: "+primitiveDouble );

System.out.println("intObj: "+intObj);
System.out.println("doubleObj: "+doubleObj);

System.out.println("Converted from String(Integer): "+number);
System.out.println("Converted from String(Double): "+decimal);

System.out.println("String value of Integer: "+str1);
System.out.println("String value of Double: "+str2);

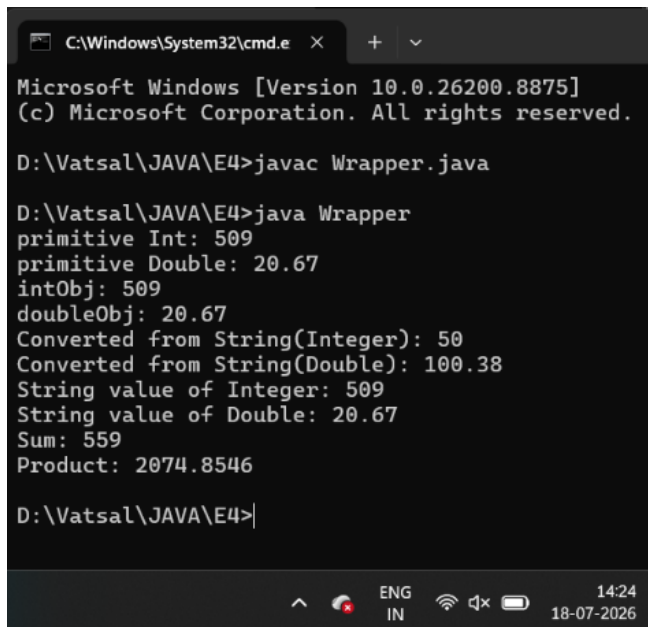
System.out.println("Sum: "+sum);
System.out.println("Product: "+product);

}

}

```

Output



```

C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

D:\Vatsal\JAVA\E4>javac Wrapper.java

D:\Vatsal\JAVA\E4>java Wrapper
primitive Int: 509
primitive Double: 20.67
intObj: 509
doubleObj: 20.67
Converted from String(Integer): 50
Converted from String(Double): 100.38
String value of Integer: 509
String value of Double: 20.67
Sum: 559
Product: 2074.8546

D:\Vatsal\JAVA\E4>

```

Exercise

1. Create a program to convert student marks from String format to Integer and calculate total marks.

Code

```
import java.util.Scanner;

public class StudentMarks {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Student Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Subject 1 Mark: ");
        String mark1 = sc.nextLine();

        System.out.print("Enter Subject 2 Mark: ");
        String mark2 = sc.nextLine();

        System.out.print("Enter Subject 3 Mark: ");
        String mark3 = sc.nextLine();

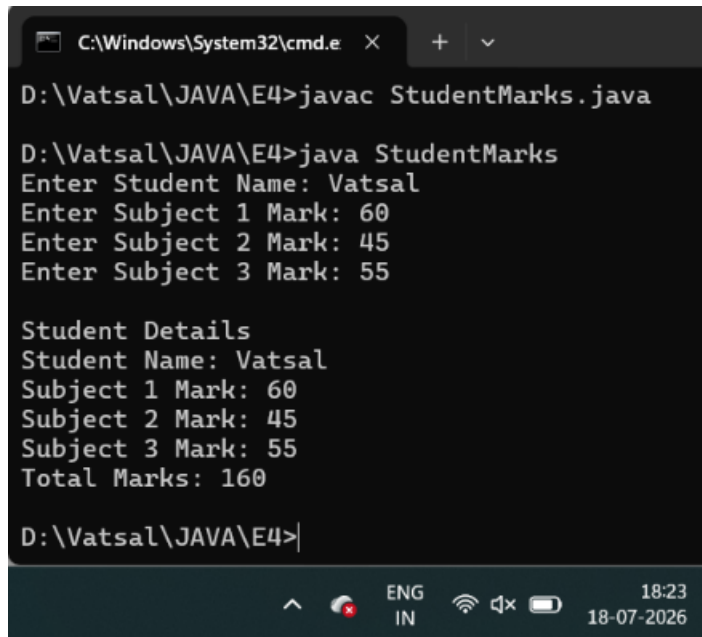
        Integer m1 = Integer.valueOf(mark1);
        Integer m2 = Integer.valueOf(mark2);
        Integer m3 = Integer.valueOf(mark3);

        int total = m1 + m2 + m3;

        System.out.println("\nStudent Details");
        System.out.println("Student Name: " + name);
        System.out.println("Subject 1 Mark: " + m1);
        System.out.println("Subject 2 Mark: " + m2);
        System.out.println("Subject 3 Mark: " + m3);
        System.out.println("Total Marks: " + total);

        sc.close();
    }
}
```

Output



```
C:\Windows\System32\cmd.e X + v
D:\Vatsal\JAVA\E4>javac StudentMarks.java
D:\Vatsal\JAVA\E4>java StudentMarks
Enter Student Name: Vatsal
Enter Subject 1 Mark: 60
Enter Subject 2 Mark: 45
Enter Subject 3 Mark: 55

Student Details
Student Name: Vatsal
Subject 1 Mark: 60
Subject 2 Mark: 45
Subject 3 Mark: 55
Total Marks: 160

D:\Vatsal\JAVA\E4>|
```

2. Develop an **Employee Payroll System** that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

Code

```
import java.util.Scanner;

public class EmployeePayroll {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Employee ID: ");
        String empId = sc.nextLine();

        System.out.print("Enter Basic Salary: ");
        String basicSalary = sc.nextLine();

        System.out.print("Enter Bonus Amount: ");
        String bonusAmount = sc.nextLine();

        Integer id = Integer.valueOf(empId);
        Double salary = Double.valueOf(basicSalary);
        Double bonus = Double.valueOf(bonusAmount);
```

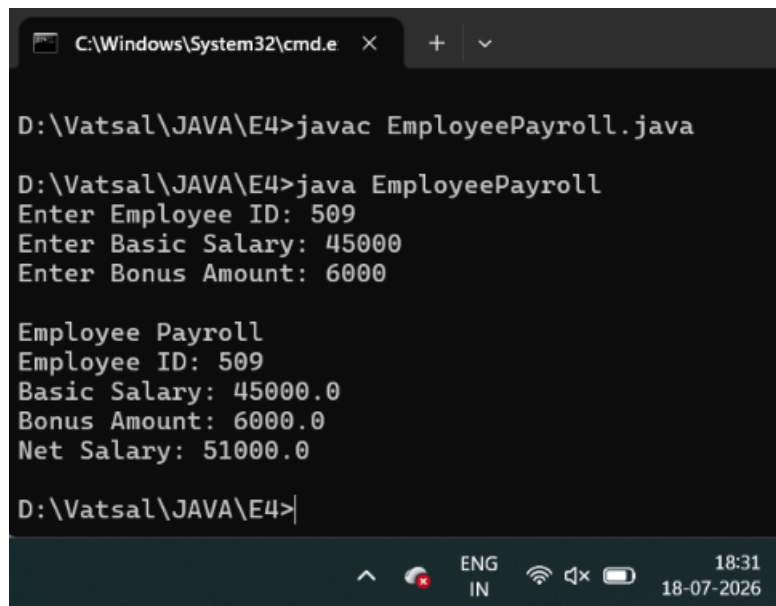
```
// Validation
if (salary <= 0) {
    System.out.println("Invalid Salary! Basic Salary must be greater than 0.");
} else if (bonus < 0) {
    System.out.println("Invalid Bonus! Bonus cannot be negative.");
} else {

    double netSalary = salary + bonus;

    System.out.println("\nEmployee Payroll");
    System.out.println("Employee ID: " + id);
    System.out.println("Basic Salary: " + salary);
    System.out.println("Bonus Amount: " + bonus);
    System.out.println("Net Salary: " + netSalary);
}

sc.close();
}
```

Output



```
C:\Windows\System32\cmd.e X + v

D:\Vatsal\JAVA\E4>javac EmployeePayroll.java

D:\Vatsal\JAVA\E4>java EmployeePayroll
Enter Employee ID: 509
Enter Basic Salary: 45000
Enter Bonus Amount: 6000

Employee Payroll
Employee ID: 509
Basic Salary: 45000.0
Bonus Amount: 6000.0
Net Salary: 51000.0

D:\Vatsal\JAVA\E4>|
```